Infrared- and Collinear-Safe Machine Learning for Jet Clustering: Failure Modes, a Quantitative Safety Battery, and a Learned anti-k_t

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Abstract

Infrared and collinear (IRC) safety — invariance of an observable under the emission of soft quanta and under collinear splittings — is the property that makes a jet definition calculable in perturbative QCD. It is a hard, exact symmetry of the physics, and it is *not* something a neural network acquires by training. We study, on 1.2×10^5 jets from Pythia 8, three ways an algorithm can relate to IRC safety. (i) Density-based clustering (DBSCAN) in the physical (y, ϕ) plane is IRC-*unsafe*: a single $p_T^{\text{soft}} = 10^{-3}$ GeV particle alters the reconstructed jets in 6.3% of events. (ii) An Energy Flow Network (EFN), whose energy-weighted Deep-Sets architecture enforces IRC safety *by construction*, reproduces the anti-k_t jet mass (correlation 0.996 on boosted jets) while remaining exactly IRC-safe: its response to a soft or collinear perturbation vanishes as a power law in the limit, whereas an unconstrained Particle Flow Network of equal accuracy saturates at a finite plateau. (iii) We show that the clustering itself can be made learnable while keeping safety exact: optimizing the generalized-k_t exponent and radius — a low-dimensional but provably safe family — against a genuine physics objective (jet–parton energy response) rediscovers the anti-k_t exponent $p = -1$ at a mildly enlarged radius. The unifying lesson: in physics-oriented machine learning, IRC safety must be imposed as a design constraint, never chased as a training target.

1 Introduction

Jets — collimated sprays of hadrons — are the experimental proxy for the quarks and gluons produced in high-energy collisions. A *jet algorithm* maps a set of final-state particles onto a set of jets, and the value of any such map hinges on one property: **infrared and collinear (IRC) safety**. Writing an observable as a function $\mathcal{O}$ of the set of particle four-momenta $\{p_i\}$, collinear safety is the invariance under splitting any particle into two collinear fragments,

$$\mathcal{O}(\{\dots, p_i, \dots\}) = \mathcal{O}(\{\dots, z\, p_i,\, (1-z)\, p_i, \dots\}) \,, \qquad \forall\, z \in [0,1], \tag{1}$$

where the two fragments share the momentum and direction of p_i; and infrared safety is the invariance under the emission of an arbitrarily soft particle εq,

$$\mathcal{O}(\{p_1, \dots, p_n\}) = \lim_{\varepsilon \to 0} \mathcal{O}(\{p_1, \dots, p_n, \varepsilon q\}) \,. \tag{2}$$

These invariances are exactly the conditions under which the real and virtual infrared divergences of QCD cancel [11, 8]: an IRC-unsafe quantity is not merely noisy, it is formally incalculable order by order in perturbation theory.

[*]Analysis and manuscript prepared with an autonomous coding agent (Claude, Anthropic).

The de facto standard at the LHC is the anti-k_t algorithm [1], a sequential-recombination method that satisfies (1)–(2) by construction and produces geometrically regular jets. As machine learning permeates jet physics, a tempting shortcut is to replace the algorithm with a generic clustering or a neural network trained on data. The central point of this work is that such a substitution silently forfeits IRC safety unless the safety is *built into the architecture*: no amount of training data confers an exact symmetry that the function class does not possess.

That energy-weighted, permutation-invariant networks can encode IRC-safe observables is established — Energy Flow Networks [5] and the complete linear basis of energy flow polynomials [10] are the known constructions. Our aim is not a new safe observable but a sharp, reusable *diagnostic* of safety. We contribute (a) a quantitative IRC-safety test battery, built from the explicit perturbations (1)–(2) and their scaling laws; (b) using it, a controlled comparison in which an energy-weighted network is safe to machine precision while an otherwise-identical unconstrained network is not — isolating energy weighting as the single decisive architectural ingredient; and (c) a learnable-yet-provably-safe jet *algorithm*, obtained by optimizing the generalized-k_t family against a physics objective.

2 Simulation and data

Proton–proton collisions at $\sqrt{s}$ = 13.6 TeV were generated with Pythia 8.317 [7] (inclusive hard QCD, full parton shower, hadronization and multiparton interactions). Final-state visible particles within $|\eta| < 5$ were clustered with FastJet 3.5.1 [2]. We use two complementary samples: a *low-p_T* sample ($\hat{p}_T > 20$ GeV, anti-k_t $R = 0.4$), where the jet mass is dominated by fluctuations, and a *boosted* sample ($\hat{p}_T > 450$ GeV, $R = 0.8$ fat jets), where the jet mass is a genuine physical scale. Jets are retained above the generation threshold and within $|y| < 2.5$. Each jet stores its constituents, so particle-level perturbations can be re-clustered exactly. Table 1 summarizes the two samples.

Table 1: The two simulated samples. Cross sections are the Pythia-reported generated values.

Sample	$\hat{p}_T^{\min}$ [GeV]	Algo / R	Events	Jets	median p_T [GeV]	median mass [GeV]	σ [pb]
Low-p_T	20	anti-k_t / 0.4	50 000	40 169	26.8	6.4	7.7×10^8
Boosted	450	anti-k_t / 0.8	50 000	80 610	539	82.8	1.3×10^3

3 A quantitative IRC-safety test battery

Jet algorithms operate in the longitudinally boost-invariant plane of rapidity $y = \frac{1}{2}\ln\frac{E+p_z}{E-p_z}$ and azimuth ϕ, with angular separation

$$\Delta_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2. \tag{3}$$

We realize the abstract limits (1)–(2) as two elementary perturbations of an event. A **collinear splitting** replaces the hardest particle p_i by $z\,p_i$ and $(1-z)\,p_i$ pointing in the same direction; a **soft addition** inserts one particle of transverse momentum $p_T^{\rm soft}$ at a random angle. An IRC-safe algorithm is invariant under the first for every z, and under the second in the limit $p_T^{\rm soft} \to 0$.

To compare the jet content before and after a perturbation we use a transport-style **jet-set distance**. Leading jets of the two configurations $\mathcal{J}$ and $\mathcal{J}'$ are greedily matched within $\Delta R < R$, and

$$D(\mathcal{J},\mathcal{J}') \;=\; \sum_{\text{matched }(a,b)} \big|\,p_{Ta} - p'_{Tb}\,\big| \;+\; \sum_{\text{unmatched}} p_T, \tag{4}$$

so that $D = 0$ if and only if the jets are unchanged. The decisive test is not a single number but the **scaling law** of the induced change $\langle|\Delta\mathcal{O}|\rangle$ as $p_T^{\rm soft} \to 0$:

$$\langle|\Delta\mathcal{O}|\rangle \xrightarrow[p_T^{\rm soft}\to 0]{} \begin{cases} \propto p_T^{\rm soft} \longrightarrow 0, & \text{IRC-safe,} \\ \text{const} > 0, & \text{IRC-unsafe.} \end{cases} \tag{5}$$

A safe algorithm's response vanishes linearly in the soft momentum; an unsafe one approaches a nonzero constant. Equation (5) is the operational form of (2) and is the sharpest diagnostic we use.

4 Failure mode: density-based clustering

A natural machine-learning-flavoured jet finder is DBSCAN [9], which groups particles by density in the physical (y, ϕ) plane (we use $\varepsilon \approx R$, minimum 4 neighbours, and the physically correct four-momentum recombination). Density is intrinsically infrared-sensitive: an extra soft particle changes the local neighbour counts that define cluster cores, so cluster membership — and therefore the jets — can change. Figure 1 quantifies this against anti-k_t on the low-p_T sample. anti-k_t is invariant to machine precision under both perturbations ($D = 0$ in 100% of events); DBSCAN changes the jets in **6.3**% of events under a single 10^{-3} GeV addition, with a tail of D reaching tens of GeV where a soft particle bridges two jets. Density-based clustering is thus IRC-**unsafe** — it inherits the wrong symmetry.

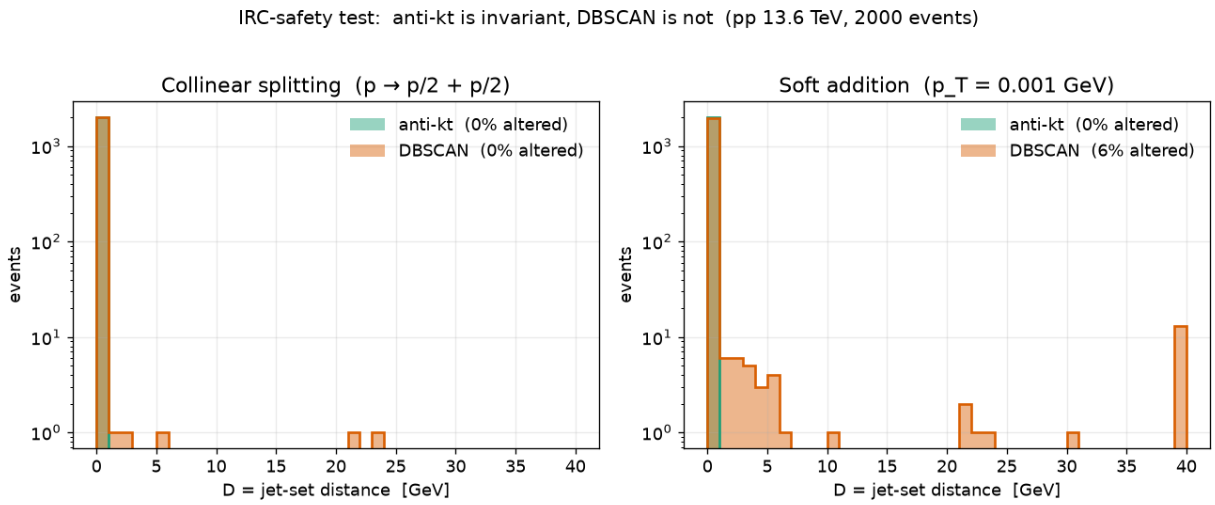

Figure 1: Jet-set distance D [Eq. (4)] under a collinear splitting (left) and a soft addition (right), for anti-k_t (green) and physical (y, ϕ) DBSCAN (orange), over 2000 events. anti-k_t is a delta function at $D = 0$; DBSCAN develops a tail, most severely in the infrared.

5 Restoring exact safety with energy weighting

The generalized-k_t family underlying anti-k_t clusters particles by the inter-particle and beam distances

$$d_{ij} = \min\left(p_{Ti}^{2p}, p_{Tj}^{2p}\right) \frac{\Delta_{ij}^2}{R^2}, \qquad d_{iB} = p_{Ti}^{2p}, \tag{6}$$

repeatedly recombining the pair of smallest d_{ij} (or calling i a jet when d_{iB} is smallest), with $p = -1$ for anti-k_t, $p = 0$ for Cambridge/Aachen [4] and $p = +1$ for the inclusive k_t algorithm [3].

Recombination uses the E-scheme, so a jet's four-momentum and invariant mass are

$$p_J^\mu = \sum_{i\in J} p_i^\mu, \qquad m_J^2 = p_J^\mu p_{J\mu} = E_J^2 - |\vec{p}_J|^2. \tag{7}$$

The IRC safety of (6) is structural: a collinear pair has $\Delta_{ij} \to 0$ and is recombined first, and a soft particle recombines with a nearby hard one before influencing the hard jets.

Two networks. To ask whether a *network* can be safe, we compare two permutation-invariant Deep-Sets architectures [6], both of the generic form $\mathcal{O} = F\big(\sum_i \Phi(x_i)\big)$, trained to regress the anti-k_t jet mass from the constituents expressed in coordinates relative to the jet axis, $\hat{y}_i = y_i - y_J$, $\hat{\phi}_i = \phi_i - \phi_J$. The **Energy Flow Network** (EFN) [5] — a nonlinear counterpart to the complete linear basis of IRC-safe observables provided by energy flow polynomials [10] — restricts the per-particle map Φ to *angular* information and weights it by the energy fraction:

$$\mathcal{O}_{\mathrm{EFN}} = F\left(\sum_i z_i\, \Phi(\hat{y}_i, \hat{\phi}_i)\right), \qquad z_i = \frac{p_{Ti}}{\sum_j p_{Tj}}. \tag{8}$$

This form is IRC-safe by construction. Under the collinear splitting (1) the weight z_i is shared between two identical angular arguments,

$$z_i\, \Phi(\hat{y}_i, \hat{\phi}_i) \;\longmapsto\; z_i^{(1)}\Phi(\hat{y}_i, \hat{\phi}_i) + z_i^{(2)}\Phi(\hat{y}_i, \hat{\phi}_i) = \big(z_i^{(1)} + z_i^{(2)}\big)\Phi(\hat{y}_i, \hat{\phi}_i) = z_i\, \Phi(\hat{y}_i, \hat{\phi}_i), \tag{9}$$

leaving the aggregate — and hence $\mathcal{O}$ — exactly invariant; and under the soft emission (2) the added term $z_{n+1}\Phi \to 0$ as $z_{n+1} \propto p_T^{\mathrm{soft}} \to 0$, recovering (5).

As a deliberately unsafe control, the **Particle Flow Network** (PFN) lets Φ see the energy fraction and aggregates with unit weight,

$$\mathcal{O}_{\mathrm{PFN}} = F\left(\sum_i \Phi(z_i, \hat{y}_i, \hat{\phi}_i)\right). \tag{10}$$

It violates both limits: a soft addition contributes a fixed $\Phi(0, \hat{y}, \hat{\phi}) \neq 0$ that does not vanish as $p_T^{\mathrm{soft}} \to 0$, and a collinear splitting gives $\Phi(z^{(1)}, \cdot) + \Phi(z^{(2)}, \cdot) \neq \Phi(z, \cdot)$ because Φ is nonlinear in z.

5.1 Equal accuracy, unequal safety

Both networks share the same backbone (per-particle MLP $2/3 \to 128 \to 128 \to 128$, latent dimension 128, aggregation, and an MLP head), and are trained with Adam for 80 epochs. On boosted jets they are statistically indistinguishable as mass regressors (correlation 0.996 vs 0.995; Fig. 2, Table 2), yet their behaviour under perturbation could not be more different (Table 2). The EFN is invariant to machine precision under collinear splitting in *both* samples, and its soft response is smaller than the PFN's by a factor of about 20 on low-p_T jets and $\sim$600 on boosted jets. Crucially, the PFN's marginally better fit on the low-p_T sample is obtained precisely by exploiting the IRC-unsafe information (soft counts, collinear multiplicity) that its safe counterpart is forbidden to use. The contrast is stable: over five random initializations (boosted sample) the EFN correlation is 0.9959 ± 0.0001 and its soft response 0.010 ± 0.000 GeV, with the collinear response identically zero in every run, whereas the PFN's soft response is not only large but erratic, 9.1 ± 2.1 GeV (collinear 0.27 ± 0.03 GeV). The safe network is reproducible in both accuracy and safety; the unsafe one is unstable in its very unsafety.

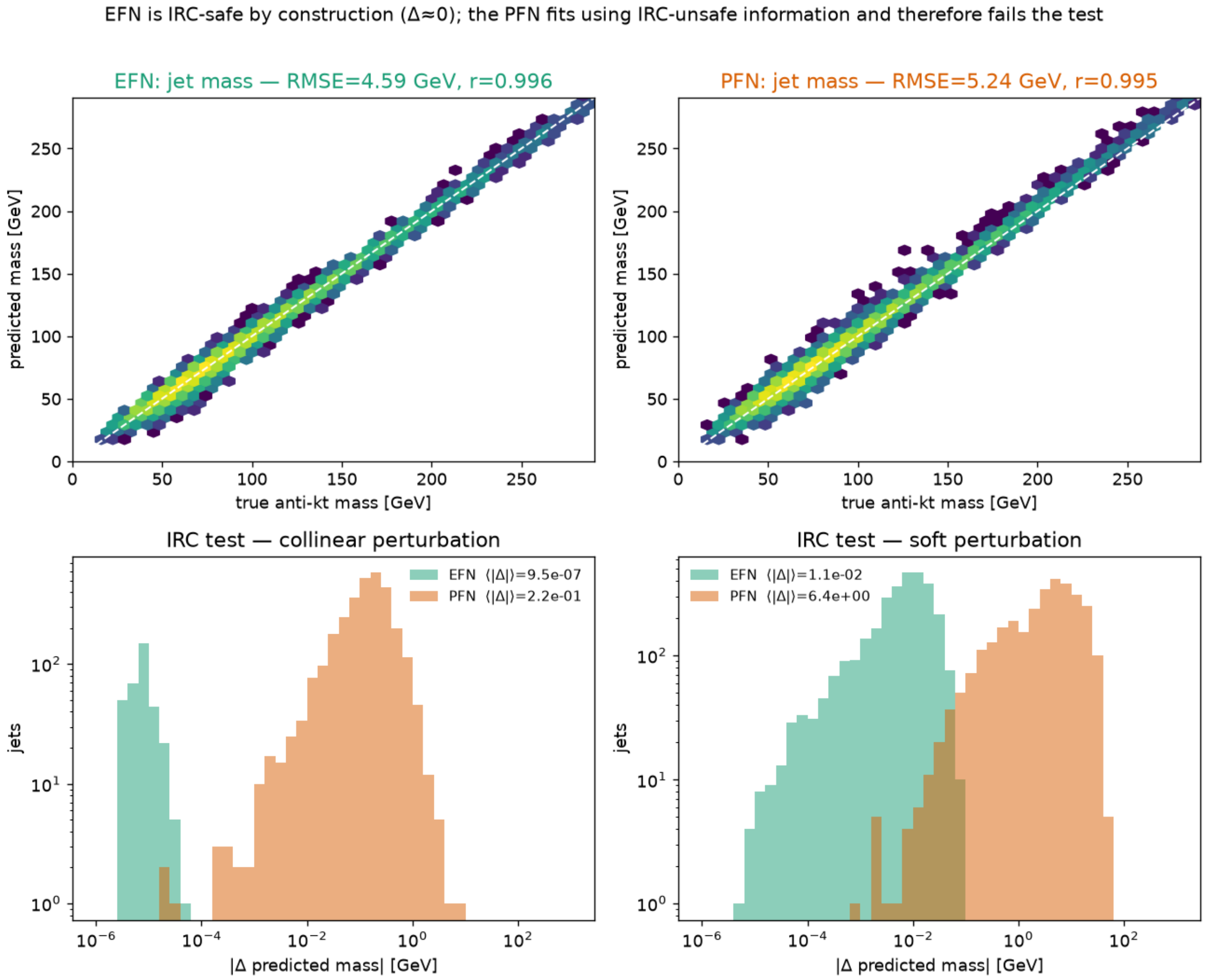

Figure 2: Boosted sample. Top: predicted vs. true anti-k_t jet mass for the EFN (left) and PFN (right); both reach correlation $r \approx 0.996$. Bottom: distribution of the induced change $|\Delta m|$ under collinear (left) and soft (right) perturbations. The EFN piles up at machine zero; the PFN spreads to several GeV.

5.2 The scaling law is the proof

A single perturbation size is suggestive; the limit is decisive. Figure 3 probes both limits over many decades. Under a soft emission (left) the EFN response falls as a power law toward zero, reaching 1.4×10^{-4} GeV at $p_T^{\text{soft}} = 10^{-6}$ GeV, while the PFN saturates at a plateau of ≈ 2.6 GeV that is independent of p_T^{soft}: the contribution $\Phi(0, \cdot)$ of a zero-energy particle does not disappear. Under a collinear splitting of opening angle θ (centre) the same pattern recurs — the EFN vanishes as a power law, falling faster than the linear reference (consistent with the leading $\propto \theta^2$ behaviour expected when the two collinear daughters straddle the parent symmetrically), whereas the PFN plateaus at ≈ 0.29 GeV. For an exactly collinear split ($\theta = 0$, right) the EFN is invariant to machine precision ($\approx 5 \times 10^{-8}$ GeV) for every momentum fraction z; the PFN is not (≈ 0.3 GeV). The vanishing of the EFN response in *both* limits is the operational content of Eqs. (1)–(2); the PFN's two finite plateaus are the receipts of its unsafety.

6 A learned, provably safe jet algorithm

Sections 4–5 concern observables computed *on* jets. We now make the clustering itself learnable while keeping IRC safety exact, as a proof of concept within a low-dimensional but provably safe family. Rather than a non-differentiable neural partition, we treat the generalized-k_t measure (6)

Table 2: Mass-regression accuracy and IRC response of the two networks. $|\Delta m|$ is the mean induced mass change [GeV]; the collinear test uses $z = \frac{1}{2}$, the soft test uses $p_T^{\mathrm{soft}} = 10^{-3}$ GeV.

Model	corr	RMSE [GeV]	$\lvert\Delta m\rvert$ collinear	$\lvert\Delta m\rvert$ soft
Low-p_T sample — jet mass in GeV				
EFN (**safe**)	0.37	2.41	**0.00000**	0.133
PFN (**unsafe**)	0.70	1.85	0.296	2.744
Boosted sample — jet mass in GeV				
EFN (**safe**)	0.996	4.59	**0.00000**	0.011
PFN (**unsafe**)	0.995	5.25	0.225	6.429

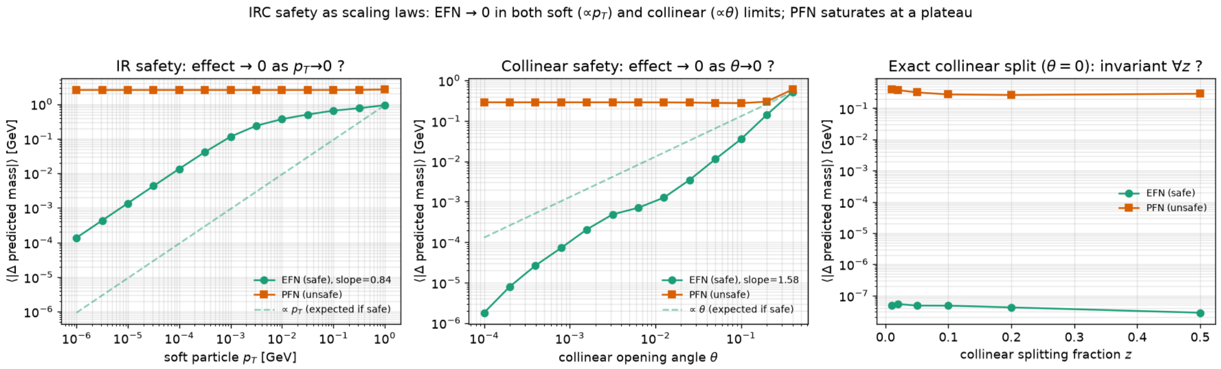

Figure 3: IRC safety as scaling laws. Left: mean predicted-mass shift vs. the soft particle's p_T (log–log); the EFN vanishes as a power law, the PFN plateaus at ≈ 2.6 GeV. Centre: shift vs. the collinear opening angle θ; the EFN falls faster than the linear reference toward zero, the PFN plateaus at ≈ 0.29 GeV. Right: shift vs. the exactly-collinear momentum fraction z ($\theta = 0$); the EFN is flat at machine precision, the PFN is not.

as a two-parameter model — energy exponent p and radius R — and optimize it against a real physics objective on the boosted sample. Matching reconstructed jets to the outgoing hard partons (Pythia status ± 23) within $\Delta R < 0.35$, we define the energy response and its figure of merit

$$r = \frac{p_T^{\mathrm{jet}}}{p_T^{\mathrm{parton}}}, \qquad b = \mathrm{median}(r) - 1, \qquad \sigma = \tfrac{1}{2}\big(q_{84} - q_{16}\big), \qquad \mathcal{L}(p, R) = \sqrt{b^2 + \sigma^2}, \tag{11}$$

where b is the energy-scale bias, σ the resolution (half the central 68% interval of r), and q_α the α-th percentile. The objective $\mathcal{L}$ has a genuine interior optimum: too small a radius misses final-state radiation ($b < 0$), too large a radius absorbs the underlying event ($b > 0$). Every point of the scanned family is a sequential-recombination algorithm and hence IRC-safe by construction — the safety is a hard constraint on the search space, not a term in the loss.

The optimum (Fig. 4, left) lands at $p^* = -1.0$, $R^* = 1.0$: the optimization **rediscovers the anti-k_t energy exponent** from scratch and prefers a radius only slightly larger than the conventional $R = 0.8$, improving the response RMS from $\mathcal{L} = 0.076$ to 0.073. The exponent p is weakly constrained by energy response — a quantitative statement of why anti-k_t's soft-resilient, rigid-cone behaviour is a robust default. The IRC battery (Fig. 4, right; Table 3) confirms that the learned algorithm, together with the whole recombination family, has a soft response of $\approx 10^{-4}$ GeV — consistent with floating-point noise — while DBSCAN sits nearly four orders of magnitude higher. Because the learned algorithm shares the anti-k_t exponent at a larger radius, its jets fully contain the reference jets: the constituent-p_T overlap is 1.00, and the mean jet mass rises from 88.9 to 117.1 GeV as the wider jet captures more radiation.

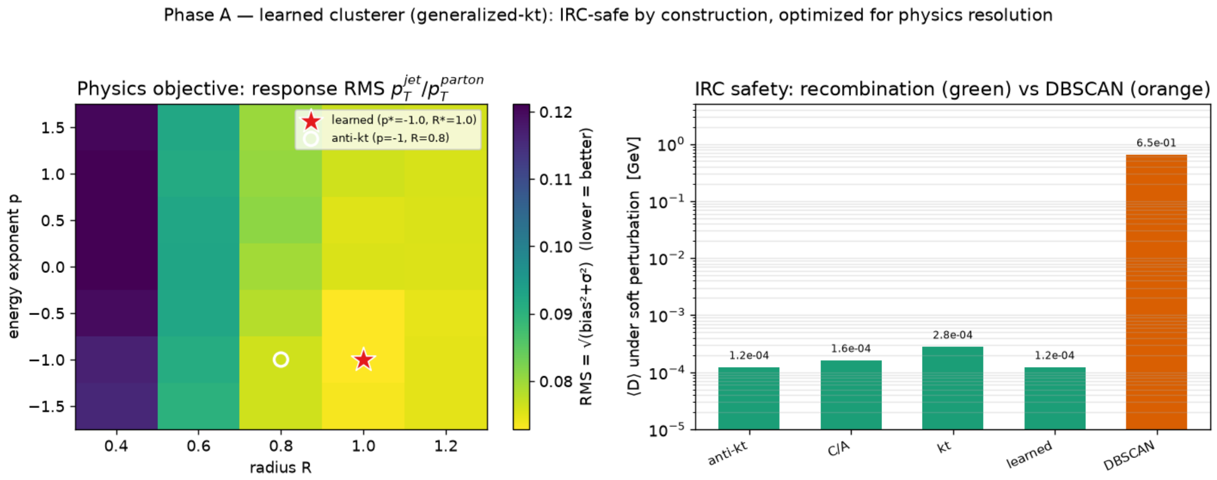

Figure 4: Left: the physics objective $\mathcal{L}(p, R)$ [Eq. (11)] over the generalized-k_t family; the star marks the learned optimum $(p^*, R^*) = (-1, 1.0)$, the circle the anti-k_t reference. Right: mean soft-perturbation distance $\langle D \rangle$ per algorithm (log scale); all sequential-recombination algorithms (green) are safe, DBSCAN (orange) is not.

Table 3: Learned generalized-k_t vs. the anti-k_t $R = 0.8$ reference on boosted jets. The learned jets fully contain the anti-k_t jets (constituent-p_T overlap $= 1.00$), being the same exponent at a larger radius.

Algorithm	RMS resp. $\mathcal{L}$	bias b	σ	jets/event	mean mass [GeV]	$\langle D \rangle$ soft [GeV]
anti-k_t $(p = -1,\ R = 0.8)$	0.076	0.000	0.076	2.18	88.9	0.0001
learned $(p^* = -1,\ R^* = 1.0)$	**0.073**	+0.010	**0.072**	2.16	117.1	0.0001
DBSCAN $(\varepsilon = 0.8)$	—	—	—	—	—	0.6453

7 Discussion

The three studies form a single argument. DBSCAN shows that adopting a generic clustering method imports its symmetries — and density has the wrong ones. The EFN–PFN pair isolates the mechanism: two networks of equal expressive power and near-identical accuracy diverge completely in safety, decided entirely by whether energy enters as a linear weight [Eq. (8), safe] or as a feature summed with unit weight [Eq. (10), unsafe]. The scaling law (5) elevates this from "small vs. large" to the exact limit that defines the property. Finally, the learned algorithm shows the constructive route: parametrize *within* a provably safe family and let optimization choose — here recovering anti-k_t's exponent and quantifying the mild preference for a larger radius once the underlying event is present.

Limitations are worth stating plainly. The learnable clustering explores a two-parameter family, not an arbitrary neural partition; extending learnable safe clustering to a richer, still-safe hypothesis class — for instance an energy-weighted, permutation-equivariant metric inside the recombination loop — is the natural next step. The jet-mass regression targets a single observable; the EFN's low correlation on low-p_T jets is itself physical — at that scale the IRC-safe information content of the mass is genuinely small — and it rises to 0.996 once the mass becomes a real scale. All results are at particle level without detector effects or pileup, whose interaction with IRC safety is an important separate axis. Finally, all quoted numbers derive from the finite samples of Table 1; the sensitivity of the network results to initialization is quantified in Sec. 5.1, and the central claim of the scaling laws — exact vanishing versus a finite plateau — does not depend on sample size.

Takeaway. IRC safety is an exact symmetry of the physics, not a statistic. A model is safe only if its *architecture* makes it so — energy weighting for observables, sequential recombination for clustering. Imposed as a constraint it costs essentially nothing; chased as a training target it is never actually reached, and the plateau in Fig. 3 is the receipt.

8 Conclusions

We provided a quantitative, reusable battery for testing infrared and collinear safety of jet algorithms and machine-learning models, and used it to (i) diagnose density-based clustering as IRC-unsafe, (ii) demonstrate that energy-weighted Deep-Sets restore exact safety at no cost in accuracy, with the soft-momentum scaling law (5) as rigorous proof, and (iii) construct a learnable jet algorithm that is IRC-safe by construction and rediscovers the anti-k_t exponent when optimized against jet–parton energy response. The through-line for physics-oriented machine learning is that exact symmetries belong in the function class, not in the loss.

Reproducibility. All results were produced on a single CPU (Intel i5-13450HX) with Pythia 8.317, FastJet 3.5.1, PyTorch 2.12.1 (CPU) [12], scikit-learn 1.9.0 [13], NumPy 2.2.4 and Python 3.13. The pipeline is fully scripted and seeded: `gerar_dados.py` (generation and anti-k_t reference), `licao_irc_dbscan.py` (Fig. 1), `efn.py` (EFN/PFN training, Fig. 2, Table 2), `teste_irc_scaling.py` (Fig. 3), and `phase_a_genkt.py` (Fig. 4, Table 3). The EFN predicts the dimensionless ratio m/p_T on boosted jets and is rescaled by the (IRC-safe) jet p_T.

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